F4, F5 and F6-IC on Koblitz and generic binary curves

Run report, ecbench, 2026-10-11 from 19:04 UTC (run run-20261011T1904Z)

1. What ran

- **Protocol.** research/f4_f5_f6_cross_family_20261011/PROTOCOL.md, committed before any measured run (commit bdd3c466).
- **Binary.** One ecbench binary, sha256 780e1084b13a5da7…, built from bdd3c466. The sessions record checkout HEAD at their start: bdd3c466 (K13), bc2d92fc (R13, K19), 83c3a2c8 (R19, X19). No file compiled into ecbench changed between those commits, and every record carries the same binary hash.
- **Host.** An x86-64 Linux cloud VM. The isolation earned per run is a mix of L1 and L2, recorded per arm in ANALYSIS.json.
- **Arms.** Four Boolean decomposers through pdp3-binary:m=3,degree=3,node_budget=8192 on binary-standard-subspace:dimension=⌈m/3⌉ (5 at m = 13, 7 at m = 19), max_trials=200000:
  - matrix F4 and matrix F5, which split on the lowest free variable;
  - inherited F4 and F6-IC, which split on the highest free variable.

  References, run on the same targets:
  - rho.negation;
  - on Koblitz curves only, rho.signed_frobenius_strong with 32 lanes, dp_bits 4.
- **Seeds and workloads.** Target seed 202610111901, measurement seed 202610111902, one round. The workloads are 8 public hash_to_subgroup_v1 targets per curve. SPEC-X19 uses the first target of each m = 19 curve.

spec	session	curve	targets	arms	verified	replay
K13	ECBS1h2b56c13eeaec	icv1-f2m13-t181-515ee569 Koblitz, r = 2,003	8	6	48/48	ok
R13	ECBS1h78657ab42aba	icv1-f2m13-tm9-6863a838 random, r = 1,367	8	5	40/40	ok
K19	ECBS1h4af82d2330b7	icv1-f2m19-t797-b6cf2467 Koblitz, r = 130,873	8	4	32/32	ok
R19	ECBS1hfff3e0643af0	icv1-f2m19-tm649-a9133fe6 random, r = 262,469	8	3	24/24	ok
X19	ECBS1h9453f8e9b41e	both m = 19 curves	1 each	2	0 / 4 (4 timeouts at the 1,800 s cap)	ok

Raw records: sessions/<spec>/records.jsonl. Replay receipts: audits/<spec>-replay-all.json. Derived values: ANALYSIS.json, written by examples/f4f5f6_cross_family_analyze.rs from the records alone.

2. Measured values

Units.

- W is counted solver word XORs per one-target run. It is measured and deterministic.
- S is charged group-addition equivalents over √r. It is measured, and for IC arms it is a lower bound, because word XORs are unpriced and F6-IC's geometric point additions are charged.
- Medians over 8 targets, with ranges, are derived by the analyser.

cell	rho-neg S	strong rho S	matrix F4 W	matrix F5 W	inherited W	F6-IC W	inherited S	F6-IC S
Koblitz 13	2.290	14.318	2.245e9	1.454e9	1.112e7	9.10e6	16.285	1,161.7
random 13	3.232	—	2.325e9	1.495e9	8.66e6	7.99e6	18.500	1,177.7
Koblitz 19	1.310	3.090	timeout	timeout	6.70e8	5.19e8	5.106	6,483.4
random 19	1.387	—	timeout	timeout	7.15e8	5.62e8	3.405	4,714.5

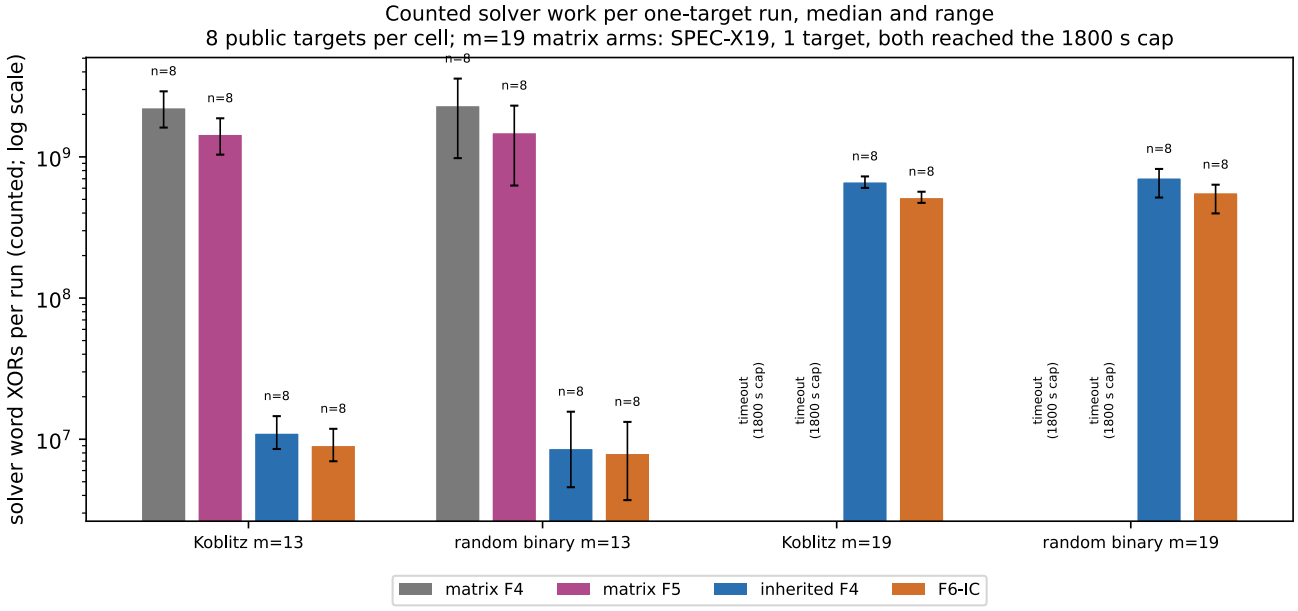


Figure 1: Counted solver word XORs per run, by engine and cell. Bars show the median and the black ticks the range over 8 targets (sessions K13, R13, K19, R19). At m = 19 the matrix arms (SPEC-X19, one target) reached the 1,800 s cap and have no bar.

3. The protocol's questions (derived; rules applied as written)

- **Q1. F5/F4 word XORs, matrix, m = 13.**
  - Koblitz: 0.6465 [0.6428, 0.6516].
  - Random: 0.6414 [0.6397, 0.6443].
  - The tree is the same on all 8 targets in both cells (equal relations, trials and reductions).
  - Rule: **F5 cuts F4's work**, in both families.
- **Q2. Matrix F4 / inherited F4 word XORs, m = 13.** 198.6 [174.5, 212.8] on Koblitz, 222.1 [173.4, 544.0] on random. No rule.
- **Q3. $\log_2$(W inherited / W F6-IC).**

cell	median [range]	rule
Koblitz 13	0.291 [0.285, 0.307]	saves
random 13	0.302 [−0.050, 0.334]	**mixed**
Koblitz 19	0.359 [0.352, 0.395]	saves
random 19	0.372 [0.319, 0.376]	saves

  - The mixed cell's one negative target ran 53 solver calls against 49: the engines returned different decompositions.
  - The per-call ratio (derived, outside the rule) is above 0 on every target of every cell, with medians 0.287, 0.306, 0.357 and 0.373.
- **Q4. IC S against same-target rho.** Every IC arm is above both rho references on all 8 targets of every cell. Rule: **IC slower than rho on every target**.
- **Q5 (X19).** All four runs (matrix F4 and F5, each m = 19 curve, one target) reached the 1,800 s cap and are recorded as timeouts, so they enter no ratio. Inherited F4 finished the same targets in about 6 s of relation phase.

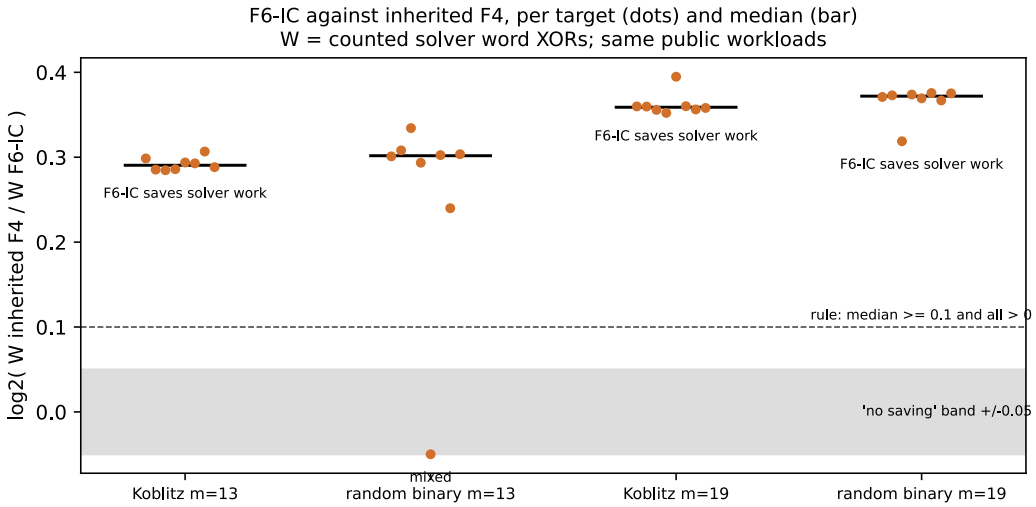


Figure 2: Per-target $\log_2$(W inherited F4 / W F6-IC) for the four cells (sessions K13, R13, K19, R19; n = 8 each). The dashed line is the preregistered 0.1 threshold, and the grey band is the "no saving" band.

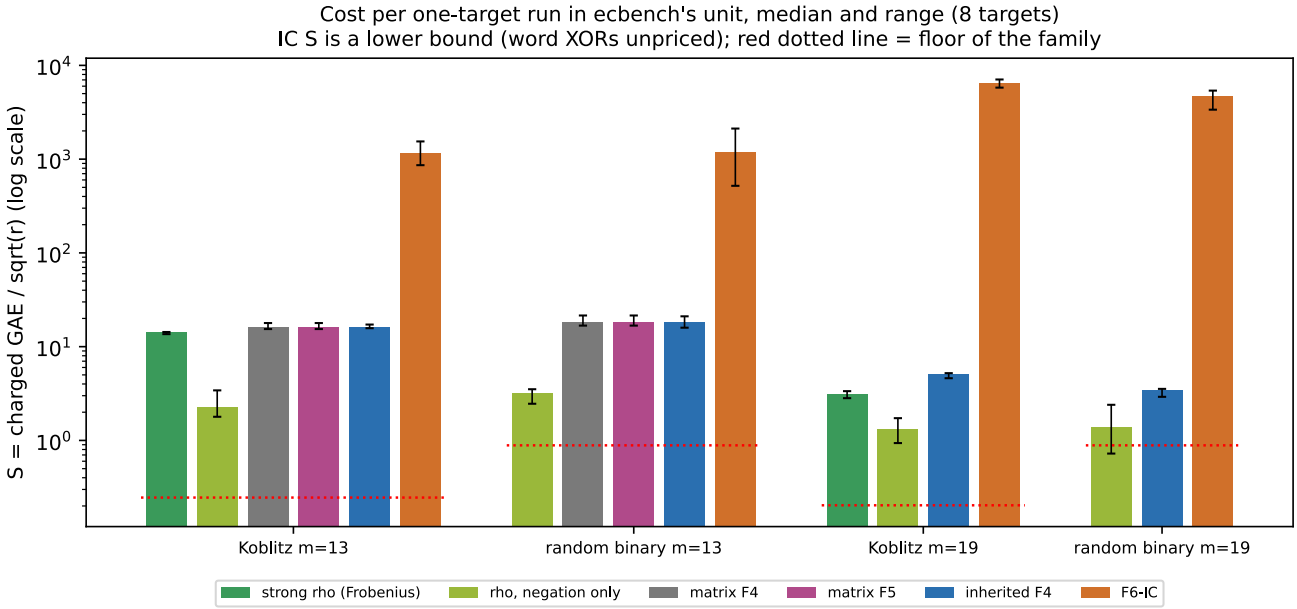


Figure 3: Lower-bound S per arm and cell, median and range over 8 targets. The red dotted line is each family's floor: $\sqrt{(\pi/4m)}$ for Koblitz, $\sqrt{(\pi/4)}$ for generic binary. Strong rho exceeds rho-negation on every Koblitz target, because its 32-lane setup dominates at these group sizes.

4. Observations and limits

- **Relation-phase wall time** (medians; mixed L1/L2, a practicality note only):

cell	matrix F4	matrix F5	inherited F4	F6-IC
Koblitz 13	30.39 s	26.08 s	0.226 s	0.112 s
random 13	32.17 s	25.36 s	0.168 s	0.096 s
Koblitz 19	> 1,800 s	> 1,800 s	6.12 s	3.63 s
random 19	> 1,800 s	> 1,800 s	6.35 s	3.59 s

- **No run failed or timed out** in K13, R13, K19 or R19. X19′s outcomes are listed above.
- **F6-IC's closure ran in every cell.** Every factor base has at most 139 points, below PR 1333′s 256-point closure cap.
- **Not shown:** any speedup in S, any exponent (two sizes), and anything about prime curves. Interpretation is in the study's `RESULT.md`.